Retrieval-Augmented Generation (RAG) Chatbot for College Regulations and Syllabus Queries

Batch No: 3 | Section: 17 | Faculty Mentor: Mrs. SD. Shareefunnisa, Assistant Professor, CSE

TEAM MEMBERS | 1. Vura Nagamani Susmitha – 251FA04D58 (Team Lead) 2. Vuyyuru Hasini – 251FA04D60 (Middle-Upper) 3. Vuyyuru Madhuri – 251FA04D62 4. Chatla Barghava Reddy – 251FA04E70

01 | PROBLEM STATEMENT & MOTIVATION

College rules (attendance, grading, CGPA, backlogs, exams, fees, hostel, library) and the syllabus of every subject are spread over long PDFs and notice boards. Students lose time finding one answer, and teachers and the college office answer the same questions every semester.

General chatbots (LLMs) do not know our college rules and may give a confident but wrong answer (hallucination). A wrong reply to "Can I write the exam with 72% attendance?" can mislead a student.

Students also type with spelling mistakes ("attendence"), short forms and different wording. Beneficiaries: students, faculty and the academic office.

Motivation: a 24×7 chatbot that gives the same, verified answer to every student and says "not found" instead of guessing.

02 | OBJECTIVES

- Build a RAG chatbot that answers college regulation and syllabus questions from a 5,000 Q&A knowledge base (CSV).
- Design a hybrid retrieval algorithm: TF-IDF word + character N-gram + MiniLM/FAISS meaning matching.
- Keep working when the question has spelling mistakes or different wording.
- Refuse unrelated questions using a confidence threshold of 0.35.
- Measure accuracy and speed and compare with simple word matching.

03 | BACKGROUND / EXISTING APPROACH

Keyword search: fast, but breaks on spelling mistakes or changed words (only 76.9% Top-1 in our test).

Only an LLM (e.g. ChatGPT): does not know college rules; may give wrong answers.

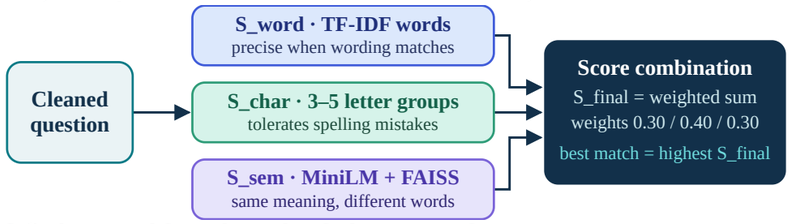
Training our own LLM: needs costly computers and retraining whenever rules change.

Our project: RAG with hybrid search – robust to typos, answers only from the CSV, and is updated just by editing the CSV.

04 | AI TECHNIQUE / ALGORITHM

Hybrid Retrieval-Based RAG Algorithm (TF-IDF + Character N-Grams + MiniLM Sentence Embeddings)

Every stored question is scored three ways, all as cosine similarity on a 0–1 scale:

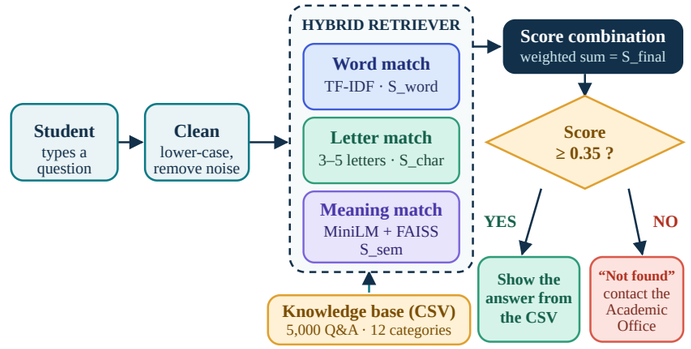


S_final = w_word·S_word + w_char·S_char + w_sem·S_sem

Best match = highest S_final (letter matching gets the greater weight). The stored answer is shown only if confidence ≥ 0.35; otherwise "not found – contact the Academic Office".

Why selected: the errors of the three methods do not coincide, so combining them gives a more reliable ranking than any single score.

05 | PROPOSED METHODOLOGY / WORKFLOW



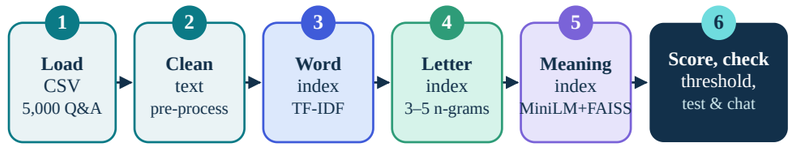
System architecture of the RAG chatbot with the hybrid retriever

06 | IMPLEMENTATION

Tools / Languages: Python, Google Colab, pandas, scikit-learn, sentence-transformers (all-MiniLM-L6-v2), FAISS, matplotlib.

Dataset / Input: CSV knowledge base of 5,000 Q&A in 12 categories (attendance, grading, CGPA, backlogs, exams, fees, hostel, library, placements, academics, project, syllabus); input is the student's typed question.

Implementation steps (classes HybridRetriever, CollegeChatbot):

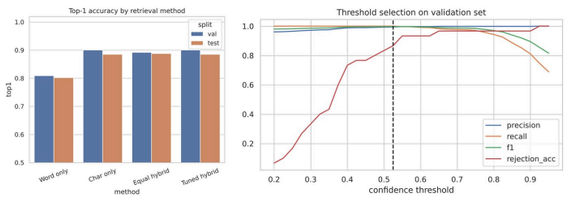


07 | RESULTS & ANALYSIS

Reported on 800 test questions (spelling mistakes + changed words) – Top-1 / Top-3 / time:

- Word matching only: 76.9% / 81.4% / 3.3 ms
- Letter matching only: 97.2% / 98.1% / 3.4 ms
- **Hybrid (our method): 97.6% / 98.4% / 3.3 ms**

Threshold 0.35: all 60 unrelated questions refused, all 300 college questions answered. 17 of 18 manual test cases correct.



Notebook graphs: Top-1 accuracy by method (left); confidence-threshold selection (right)

Key observation: letter matching fixes most typos and the hybrid is best overall; errors remain when two questions are almost identical (72% vs 75%) or a short form such as "DBMS" is not in the data.

08 | REAL-WORLD APPLICATIONS

- Student help desk: 24×7 chatbot on the college website or WhatsApp
- Examination cell and fee office: hall ticket, revaluation, supplementary exams, fees
- Teachers: quick look-up of syllabus, credits and exam pattern
- Other colleges and companies: replace the CSV (HR and customer-support FAQs)

09 | STRENGTHS, LIMITATIONS & ASSUMPTIONS

Strengths: accurate with typos, no made-up answers, about 3 ms per question, easy to update (edit the CSV), runs on free tools.

Limitations: answers only what is in the CSV; English only; no memory of earlier questions; a wrong match scoring above 0.35 is not caught; short forms must be in the dataset; MiniLM model needs a first-run download.

Assumptions: the knowledge base is correct and up to date; the CSV is sample data and must be replaced by the official regulations before real use.

10 | CONCLUSION & FUTURE SCOPE

We built a RAG chatbot whose hybrid retrieval (word + letter + meaning matching, threshold 0.35) answers college rule and syllabus questions only from verified data. It reached 97.6% Top-1 accuracy on spelling-mistake questions against 76.9% for word matching, and refused all unrelated questions.

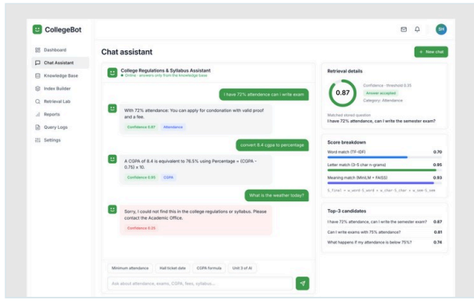
Future scope: read official regulation PDFs directly, use an LLM to phrase answers, Telugu/Hindi and voice input, conversation memory, subject short forms, deployment on the website/WhatsApp, and personal answers via the college database.

11 | REFERENCES & ACKNOWLEDGEMENTS

[1] P. Lewis et al., Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks, NeurIPS, 2020.
[2] A. Vaswani et al., Attention Is All You Need, NeurIPS, 2017.
[3] N. Reimers and I. Gurevych, Sentence-BERT, EMNLP, 2019.
[4] J. Johnson, M. Douze, H. Jégou, Billion-scale similarity search with GPUs (FAISS), IEEE Trans. Big Data, 2019.
Thanks to Mrs. SD. Shareefunnisa; scikit-learn, sentence-transformers, FAISS and Google Colab.

PROJECT DEMO

Demo: CollegeBot web UI (chat, retrieval lab, accuracy report).



CollegeBot chat assistant with retrieval details (UI screen 03)